

Organic, conventional and hydroponic vegetables: Can ^{15}N natural abundance of farm N inputs differentiate mode of production?

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ABSTRACT

Organic and synthetic N-fertilizers normally show quite different $\delta^{15}\text{N}$ values, which can be imprinted in agricultural soils and growing plants. Therefore, $\delta^{15}\text{N}$ values have been considered as a promising marker to differentiate organic plant products from other farming systems that do not exclude synthetic N-fertilizer use. The aim of this study was to evaluate the $\delta^{15}\text{N}$ values of fertilizer N sources from different farming systems and the $\delta^{15}\text{N}$ values of the plant products produced *in situ*. Samples of soil, substrates, N sources, and plant products (lettuce, tomato, and strawberry) were sampled from well documented organic (n = 2), conventional (n = 4) and hydroponic (n = 2) farms in the southern Brazil. Plant parts were also compared. $\delta^{15}\text{N}$ values of N-inputs (substrates, organic and synthetic fertilizers) overlapped, within the range of -2 to $+8$ ‰. Soil samples from conventional farms ($+5.3$ to $+9.8$ ‰) showed higher $\delta^{15}\text{N}$ values than from organic systems ($+4.9$ to $+6.5$ ‰). As a consequence, $\delta^{15}\text{N}$ values had limited value as a tool to differentiate plant products from different farming systems. However, it may be a potentially more useful technique for strawberries but not for tomatoes and lettuce. The inconsistency found in the differentiation between farming systems can be attributed to the overlapping of $\delta^{15}\text{N}$ values of the fertilizer N sources used on those farms.

1. Introduction

Stable isotopes ratios of $^2\text{H}/^1\text{H}$, $^{13}\text{C}/^{12}\text{C}$, $^{15}\text{N}/^{14}\text{N}$, $^{18}\text{O}/^{16}\text{O}$ and $^{34}\text{S}/^{32}\text{S}$ have been used for authentication of the geographical origin of food and to differentiate organic and conventional plant and animal products (Danezis et al., 2016; Inácio et al., 2015a; Inácio and Chalk, 2017). N stable isotope ratio ($^{15}\text{N}/^{14}\text{N}$) analysis at natural abundance ($\delta^{15}\text{N}$) has been suggested as a potential tool to differentiate organic and conventional plant products due to its relationship to the N fertilizer source used to supply crop nutrients for the crop. In fact, fertilizer N sources influence $\delta^{15}\text{N}$ signatures across the soil-plant-groundwater continuum in the agricultural landscape (Choi et al., 2017).

Synthetic N fertilizers (ammonium salts, nitrate, and urea) are derived from ammonia (NH_3) produced by the Haber-Bosch process, which involves the catalytic reduction of atmospheric N_2 at high

temperature and pressure by H_2 derived from methane or natural gas, a process that does not cause N isotope fractionation. Therefore, the $\delta^{15}\text{N}$ signatures of synthetic fertilizers are expected to be close to that of atmospheric N_2 ($\delta^{15}\text{N} = 0$ ‰ by definition). Meanwhile, organic N fertilizers are generally naturally enriched (positive values) in ^{15}N compared to synthetic N fertilizers. Choi et al. (2017) reports synthetic N fertilizers have slightly positive or negative $\delta^{15}\text{N}$ values within the range of -8 to $+6.2$ ‰, with 90 % of samples ranging from -3 to $+3$ ‰, while manures show more positive values in the range of $+2.7$ to $+16.2$ ‰, with 90 % of values within $+3$ and $+14$ ‰. Composted manures range from $+4.9$ to $+20.9$ ‰, with 90 % of samples ranging from $+8$ to $+20$ ‰.

These variations in $\delta^{15}\text{N}$ values are due to isotopic fractionation processes, especially NH_3 volatilization, nitrification and dissimilatory NO_3^- reduction (biological denitrification), that are the major N

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transformations affecting the natural ^{15}N isotopic composition of animal excreta and composts (organic fertilizers) (Chalk et al., 2014). Frequently, NH_3 volatilization is considered the most relevant fractionation process because of its high loss and high fractionation factor (Cejudo and Schiff, 2018). Ammonia volatilized is very depleted in ^{15}N (e.g. -31‰ to -15‰ , Hristov et al. (2009)), resulting in an enrichment of the remaining N in manure and compost (Lee et al., 2011). A highly significant positive linear relationship was observed between cumulative NH_3 loss and the $\delta^{15}\text{N}$ signature of stored manure (Hristov et al., 2009). Similarly, NH_3 volatilization and nitrification in the early stages of composting, and loss of NO_3^- in the later stage through denitrification are the primary reasons for the increase in ^{15}N in composted manure (Kim et al., 2008). Thus, the increase in $\delta^{15}\text{N}$ values of total and inorganic N during manure storage and composting may indicate substantial N losses (Inácio et al., 2018).

The $\delta^{15}\text{N}$ signatures of synthetic fertilizers, manure, composted manure, and non-manure N sources (e.g. cover crops, crop residues) might be a stronger factor causing variations in the $\delta^{15}\text{N}$ values of soils at the farm scale than natural factors such as precipitation and temperature, which would be dominant factors at a large scale in non-cultivated soils (Choi et al., 2017; Martinelli et al., 1999). In agricultural soils the ^{15}N of bulk soils is often higher than those in natural undisturbed soils due to the input of anthropogenic N sources and subsequent loss of ^{15}N -depleted inorganic N as well as crop harvesting which removes ^{15}N -depleted biomass. N loss processes lead to ^{15}N -enrichment of the remaining NH_4^+ in soil. The $\delta^{15}\text{N}$ of crops is linked primarily to the soil but the range of crops is wider (-2 to $+21\text{‰}$) than that of soil ($+1$ to $+12\text{‰}$) (Choi et al., 2017). In fact, plants take up a small fraction of the bulk soil N, the inorganic N pool in the soil solution, which has the $\delta^{15}\text{N}$ value influenced by the isotopic fractionation linked to those N transformation processes in soil.

Crop $\delta^{15}\text{N}$ values well below the value for soil organic N is a general indication that synthetic N fertilizers have been applied, while values well above may indicate manure or composted manure additions (Bateman et al., 2005; Choi et al., 2017; Inácio et al., 2015b; Nakano et al., 2003). The extent of these differences will depend on the fertilizer $\delta^{15}\text{N}$ value, and the rates of application over a given period of time. In addition, the availability of N from an organic source plays a greater role in determining crop $\delta^{15}\text{N}$ values than the $\delta^{15}\text{N}$ value of the N source itself (Choi et al., 2006). $\delta^{15}\text{N}$ values of grown plants are strongly influenced by inorganic N (NH_4^+ and NO_3^-) available in the soil solution that comes from fertilizers and soil organic matter mineralization (Dawson et al., 2002; Yun and Ro, 2008) and the higher the N availability of the source applied (e.g. synthetic fertilizers), the stronger the effect of its ^{15}N isotope composition on the crop $\delta^{15}\text{N}$ values (Choi et al., 2017; Inácio et al., 2015b).

Therefore, $\delta^{15}\text{N}$ composition appears to be a promising marker to distinguish organically- and conventionally (non-organically)-fertilized plant products. However, the use of an organic N fertilizer source does not denote that the product derives exclusively from an organic system, whereas use of a synthetic fertilizer implies a non-organic system. As shown in the data compilation made by Inácio et al. (2015a), the differences in $\delta^{15}\text{N}$ values are large and statistically significant for most vegetables and annual crops, and for faster growing vegetables (e.g. maturity < 80 days, Rogers (2008)) than for slower growing vegetables. Larger differences in $\delta^{15}\text{N}$ values between organic and conventional vegetables was found for tomatoes and Chinese cabbage, whereas smaller differences have been reported for lettuce and carrots (Inácio et al., 2015a). For example, larger differences in $\delta^{15}\text{N}$ values were found between organic and conventional tomatoes ($\sim 8.6\text{‰}$) compared to lettuce ($\sim 3.3\text{‰}$) and carrots ($\sim 3.2\text{‰}$), the latter not statistically significant (Bateman et al., 2005). Perennial or semi-perennial crops such as oranges, peaches and strawberries tend to show smaller differences in $\delta^{15}\text{N}$ values between organic and conventional production, but nevertheless they were statistically significant e.g. Camin et al. (2011). However, differentiation of legume products on the basis of

$\delta^{15}\text{N}$ composition is not possible due to biological N_2 fixation (e.g. soybean (Oberson et al., 2007)). In addition, when the difference in $\delta^{15}\text{N}$ between organic and synthetic N fertilizer applied to soil was $< 9.1\text{‰}$, no statistically significant differences were found between organic and conventional lettuce (Šturm et al., 2011). Thus, such a situation might result in overlapping $\delta^{15}\text{N}$ values for other vegetables, and also fraudulent use of synthetic N fertilizer by organic farmers might not be easily traced. In fact, with retail samples, comparison is confounded by potential differences in $\delta^{15}\text{N}$ due to geographic origin caused by differences in mean annual precipitation and/or temperature as well as fraud in labeling (Bateman et al., 2007; Rogers, 2008). In addition, the $\delta^{15}\text{N}$ value of plants is also affected by internal fractionation between plant parts (transport of N to different plant organs) (Inácio et al., 2015a; Wang et al., 2016).

The aim of this study was to evaluate the $\delta^{15}\text{N}$ values of N sources from different farming systems and the $\delta^{15}\text{N}$ values of the plant products produced *in situ*. Farms were located in Rio Grande do Sul State, in Brazil, one of the most representative regions of organic agriculture in the country. Samples of all N sources for cropping in each farm were collected. Lettuce, strawberries and tomatoes produced under organic, conventional and hydroponic cropping systems were sampled on these farms.

2. Materials and methods

Samples of lettuce (*Lactuca sativa* L., cultivar 'Veneranda'), tomatoes (*Solanum lycopersicum* L., cultivar 'Absoluto') and strawberry (*Fragaria × ananassa* D., cultivar 'San Andreas') were taken from three cropping systems: organic, conventional and hydroponic in Rio Grande do Sul State, Brazil ($28^\circ 08' 55''$ to $29^\circ 41' 03''$ S and $53^\circ 48' 25''$ to $54^\circ 44' 14''$ W), from September 2016 to April 2017 (Table 1). Hydroponic systems include all systems without soil, and may include sand or an organic substrate as the rooting medium with nutrients supplied by fertigation. The climate is humid subtropical (Cfa, Köppen-Geiger classification system), with mean annual temperatures ranging from $18.6\text{--}20.7^\circ\text{C}$ and annual precipitation from 1,516 to 1,842 mm National Meteorological Agency, <http://www.inmet.gov.br>).

2.1. Sampling and sample preparation

Each field crop was divided into three blocks for plant and soil sampling. Organic and synthetic fertilizers, substrates and other N inputs used were also sampled. Composite samples were formed for plant products and plant parts. For lettuce, subsamples of inner, middle and outer leaves were taken at harvest and they were mixed to make the composite sample for analysis. For tomatoes, lower, middle and higher leaves, and petioles were subsampled. Tomato fruits were sampled at two stages: unripe fruit – complete physiological maturation and 100 % green-colored fruit and ripe fruit – 60–90 % color reddish, rosy, or willowish (Alvarenga and Souza, 2004). For strawberries, leaves and petioles were subsampled and fruits were sampled at two stages: unripe fruit – green-colored fruit (when the achenes are clearly visible in the tissue of the receptacle); ripe fruit – red-colored fruit (Meier et al., 1994). Plant and fertilizer samples were dried at 60°C during 72 h in a forced-draft oven and then homogenized and ground to a fine powder. Soil samples were dried at 40°C before grinding.

2.2. Isotope-ratio analysis

Dried and milled samples were weighed into tin capsules to give $70\text{ }\mu\text{g N}$ ($\pm 5\text{ }\%$) for isotope-ratio analysis ($^{15}\text{N}/^{14}\text{N}$), using a mass spectrometer (Delta V Advantage, Thermo Fisher Scientific, Bremen, Germany) coupled with an elemental analyzer (Flash EA 2000, Thermo Fisher Scientific, Bremen, Germany). Isotope delta (δ) was expressed in relative values (‰ or per mil) according to Eq. 1.

Table 1
Sites, crops, cropping systems and sampling times.

Site	Crop	Cropping System	Abreviation	Sampling time
Agudo	Strawberry	Organic	Org	January/2017
	Lettuce	Organic	Org	December/2016 to January/2017
Cerro Largo	Tomato	Hydroponic ^a	Hydrop	October/2016
	Strawberry	Hydroponic ^a	Hydrop B	October/2016
	Lettuce	Hydroponic ^a	Hydrop B	October/2016
Ijuí	Tomato	Conventional	Conv C	September/2016 to November/2016
	Strawberry	Conventional	Conv A	September/2016 to November/2016
		Conventional	Conv B	February/2017
	Lettuce	Conventional	Conv B	September/2016 to November/2016
Santiago	Tomato	Organic	Org	October/2016 to January/2017
Santa Maria	Tomato	Conventional	Conv A	August to December/2016
	Lettuce	Hydroponic ^b	Hydrop A	February/2017
		Conventional	Conv A	November to December/2016
Três Barras	Tomato	Conventional	Conv B	October/2016 to February/2017
	Lettuce	Conventional	Conv C	October/2016 to February/2017
Boca do Monte	Strawberry	Hydroponic ^a	Hydrop A	September/2016

Note: Organic cropping systems do not use synthetic fertilizers which are used in conventional and hydroponic systems.

^a Organic substrate + fertigation.

^b Sand + fertigation.

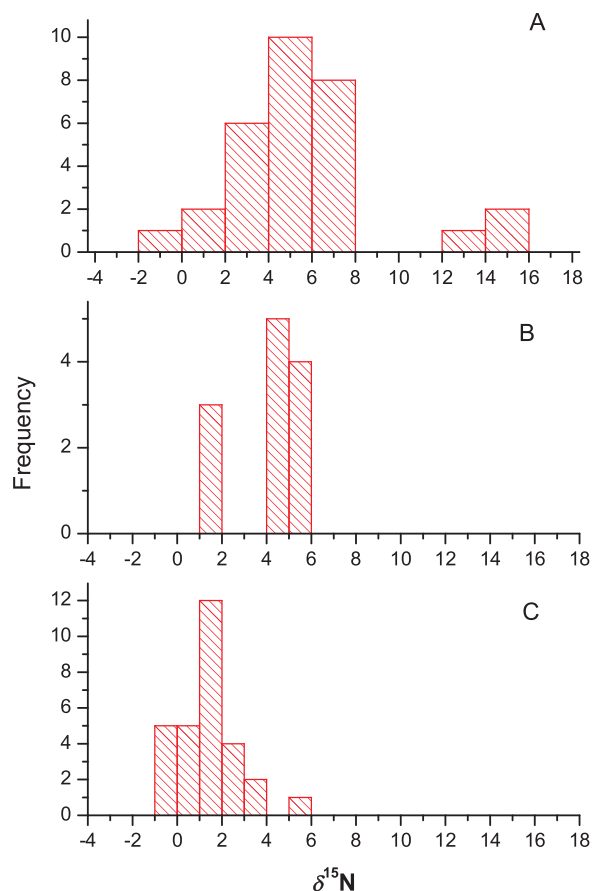


Fig. 1. Histograms of $\delta^{15}\text{N}$ values (‰) of organic inputs (A), substrates (B) and synthetic N-fertilizers (C).

$$\delta^{15}\text{N} = \frac{R\left(\frac{^{15}\text{N}}{^{14}\text{N}}\right)_{\text{sample}}}{R\left(\frac{^{15}\text{N}}{^{14}\text{N}}\right)_{\text{standard}}} - 1 \times 1000 \quad (1)$$

where $R(^{15}\text{N} / ^{14}\text{N})$ is the absolute isotope ratio according to Eq. 2.

$$R(^{15}\text{N}/^{14}\text{N}) = \frac{x(^{15}\text{N})}{1 - x(^{15}\text{N})} \quad (2)$$

and $x(^{15}\text{N})$ is the atom fraction ^{15}N .

The international standard for N stable isotope is atmospheric N_2 with an isotope ratio of $R(^{15}\text{N}/^{14}\text{N}) = 0.003\,676\,47$ (Chalk et al., 2015), and $\delta^{15}\text{N} = 0$ ‰, by definition. The δ -value can be either negative or positive depending on whether it is depleted or enriched in ^{15}N relative to the standard (Coplen et al., 2002).

2.3. Statistical analysis

Data are presented as means $\pm$ standard deviation (SD). Data were tested for normal distribution (Anderson–Darling test) and equality of variances (Barnett and Levene tests). The Dunnnett-test was used to check significant differences between means (δ -values) of cropping systems, with the organic system as the control. The significance level chosen was $\alpha = 0.05$. We used software Minitab 17 (Minitab Inc., Pennsylvania, PA, USA) for statistical analysis.

3. Results and discussion

3.1. $\delta^{15}\text{N}$ abundance in soil

Soil samples from conventional systems (+5.3 to +9.8 ‰) showed higher $\delta^{15}\text{N}$ values than those from organic systems (+4.9 to +6.5 ‰). In general, animal manure and composted manure are more often applied, and in greater amounts, in organic farming than in conventional systems. These organic inputs imprint more positive $\delta^{15}\text{N}$ values in soil than synthetic N-fertilizers (Celano et al., 2012; Choi et al., 2003). Nevertheless, the records of conventional farms in this study showed that animal manure and compost were used in previous years, which may explain the overlapping $\delta^{15}\text{N}$ values of soil samples from conventional and organic farms.

3.2. N sources

There was a relevant overlapping of $\delta^{15}\text{N}$ values of N inputs used by farmers, especially between -2 to $+8$ ‰ (Fig. 1). Only compost samples (Table 2) with a high $\delta^{15}\text{N}$ mean ($+14.3 \pm 0.3$ ‰) appear detached from the other N inputs (Fig. 1). Some samples of N inputs were not available, but NPK compound fertilizers and cattle manure commonly show $\delta^{15}\text{N}$ values in a range from -2.2 to $+2.5$ ‰ and from $+3.5$ to $+16.2$ ‰, respectively (Inácio et al., 2015a). The $\delta^{15}\text{N}$ values of every input available for each selected farm are shown in Tables 2–4.

Table 2

Location, cropping system and N sources of tomato farms with respective isotope nitrogen natural abundance.

Farm	Site	Farming System	N sources	$\delta^{15}\text{N} / \text{‰}^a$
Hydrop	Cerro Largo	Hydroponic	MAP ^c	+2.0 ± 0.1
			KNO ₃	+1.1 ± 0.4
			Substrate	+4.8 ± 0.4
Conv A	Santa Maria	Conventional	Urea	-0.2 ± 0.4
Conv B	Três Barras	Conventional	Soil	+5.6 ± 0.3
			NPK (5-20-20)	+1.9
			Compost	+14.3 ± 0.3
Conv C	Ijuí	Conventional	Soil	+8.6 ± 0.1
			NPK (4-14-8)	^b
			Cattle manure	^b
Org	Santiago	Organic	Soil	+9.7 ± 0.1
			Commercial vermicompost	+6.8 ± 0.1
			On-farm vermicompost	+3.7 ± 0.1
			Mulching	+2.8 ± 0.2
			Soil	+5.0 ± 0.1

^aMean ± standard deviation (n = 3).

^bSample not available.

^cMono ammonium phosphate.

Table 3

Location, cropping system and N sources of strawberry farms with respective isotope nitrogen natural abundance.

Farm	Site	Cropping System	N sources	$\delta^{15}\text{N} / \text{‰}^a$
Hydrop A	Boca do Monte	Hydroponic	MAP ^c	+2.5 ± 0.1
			KNO ₃	+2.3 ± 0.7
			Organic substrate	+1.3 ± 0.4
Hydrop B	Cerro Largo	Hydroponic	MAP	+1.9 ± 0.1
			KNO ₃	+1.1 ± 0.4
			Substrate	+5.7 ± 0.1
Conv A	Ijuí	Conventional	NPK (5-20-20)	^b
Conv B	Ijuí	Conventional	Soil	+7.1 ± 0.5
			NPK (6-12-36)	+3.1 ± 2.7
			Cattle manure	^b
Org	Agudo	Organic	Soil	+7.8 ± 0.0
			Poultry manure	+6.2 ± 0.7
			Rice rusk (charred)	+0.5 ± 0.5
			Substrate	+7.4 ± 0.6

^aMean ± standard deviation (n = 3).

^bSample not available.

^cMono ammonium phosphate.

3.3. $\delta^{15}\text{N}$ abundance in tomatoes, strawberries and lettuce

$\delta^{15}\text{N}$ means of organic tomatoes did not differ from those of hydroponic systems, but differed from the conventional system ($\alpha = 0.05$, Dunnett test) (Fig. 2). The probability of overlapping between organic and soilless $\delta^{15}\text{N}$ values was calculated as 83 %, and was 0 %, 0 % and 1.3 % between samples of organic and conventional systems A, B and C, respectively. Surprisingly, one conventional system (Conv B) showed samples with high $\delta^{15}\text{N}$ values (Fig. 2), an evidence of the compost as the N source (Table 2).

Also, the organic system showed the lowest $\delta^{15}\text{N}$ mean reflecting the low $\delta^{15}\text{N}$ values of N inputs, i.e. on-farm vermicompost and mulching (Table 2). In contrast, published data show a significant difference between organic and conventional tomatoes (farm and retail sampling), with $\delta^{15}\text{N}$ values in the range of +6.9 ‰ to +9.3 ‰ and from -0.9 to +0.0 ‰, respectively (Bateman et al., 2007; Rogers, 2008; Schmidt et al., 2005).

The ^{15}N natural abundance of N sources (including soil-N) on those farms overlapped greatly, except for compost (Table 2). Also, tomatoes

Table 4

Location, cropping system and N sources of lettuce farms with respective isotope nitrogen natural abundance.

Farm	Site	Farming System	N sources	$\delta^{15}\text{N} / \text{‰}^a$
Hydrop A	Santa Maria	Hydroponic	MAP ^c	+1.5 ± 0.1
Hydrop B	Cerro Largo	Hydroponic	KNO ₃	^b
			MAP	+1.9 ± 0.1
			KNO ₃	+1.1 ± 0.4
Conv A	Santa Maria	Conventional	Organic substrate	+5.0 ± 0.0
			Urea	-0.3 ± 0.1
			MAP	+1.5 ± 0.1
Conv B	Ijuí	Conventional	Soil	+5.1 ± 0.2
			NPK (5-20-20)	^b
			Soil	+7.2 ± 0.3
Conv C	Três Barras	Conventional	NPK (5-20-20)	+1.9
			Compost	+14.3 ± 0.3
			Mulching	+5.3 ± 0.2
			Soil	+5.56 ± 0.29
			Compost	+5.0 ± 0.0
Org	Agudo	Organic	Rice rusk (charred)	+5.7 ± 0.4
			Cattle manure	^b
			Soil	+6.3 ± 0.2

^aMean ± standard deviation (n = 3).

^bSample not available.

^cMono ammonium phosphate.

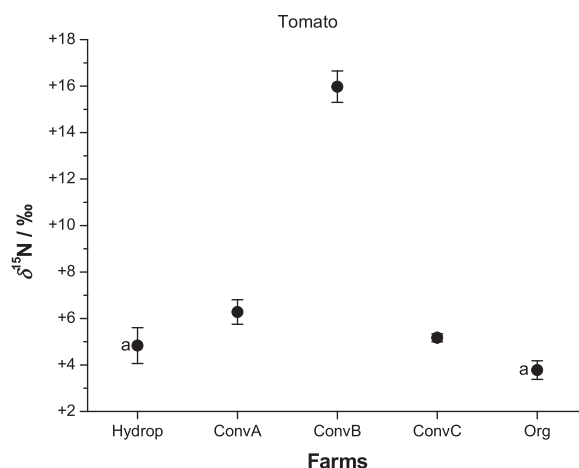


Fig. 2. $\delta^{15}\text{N}$ values (‰) of tomatoes from different cropping systems in Southern Brazil. Hydrop = Fertilization + organic substrate; Conv A = synthetic fertilizer only; Conv B and C = synthetic + organic fertilizers; Org = organic fertilizers only. Data are presented as means and standard deviations (n = 3). Same letter (a) represents no significant difference against Org mean (Dunnett-test, $\alpha = 0.05$).

$\delta^{15}\text{N}$ values did not match the arithmetic mean of N sources available on each farm, but more likely expressed one source more than the others in use, according to the N dose and the potential N recovery by the plant (Inácio et al., 2015b). The synthetic N-fertilizers did not similarly influence the $\delta^{15}\text{N}$ of tomatoes in conventional systems, despite their low $\delta^{15}\text{N}$ values and high solubility.

The strawberry organic system did not differ from conventional system A (Conv A), which showed the highest $\delta^{15}\text{N}$ mean, but was found to be statistically different ($\alpha = 0.05$, Dunnett test) from the other conventional and hydroponic systems (Fig. 3). The probability of $\delta^{15}\text{N}$ values overlapping between organic and Conv A samples was calculated as 20 %. Published data show an expected overlapping of $\delta^{15}\text{N}$ of organic and conventional strawberries. Camin et al. (2011) found $\delta^{15}\text{N}$ means of organic and conventional strawberries (farm sampling) differed statistically ($P < 0.001$), with values in the range of

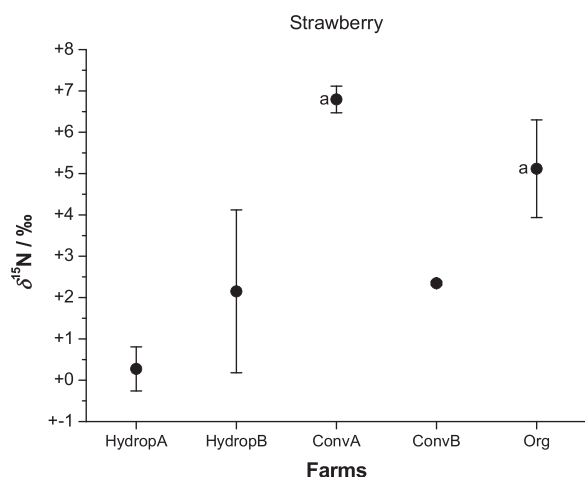


Fig. 3. $\delta^{15}\text{N}$ values (‰) of strawberries from different cropping systems in Southern Brazil. Hydrop A and B = Organic substrate + fertigation; Conv A = synthetic fertilizer only; Conv B = synthetic + organic fertilizers; Org = organic fertilizers only. Data are presented as means and standard deviations ($n = 3$). Same letter (a) represents no significant difference against Org mean (Dunnnett-test, $\alpha = 0.05$).

+1.2 to +9.8 and +2.9 to +4.0 ‰, respectively.

Considering the N sources (Table 3), it would be unlikely that the Conv A system showed high $\delta^{15}\text{N}$ values of strawberry samples. Synthetic N-fertilizer was the single N input in this farm, but strawberry samples showed the $\delta^{15}\text{N}$ mean close to the soil $\delta^{15}\text{N}$ values. Hypothetically, this may be an effect of very low synthetic N-fertilizer application rates (kg ha^{-1}). In contrast, Conv B showed a clear influence of synthetic N-fertilizer as the N source on plant $\delta^{15}\text{N}$ values.

The lettuce organic system did not differ from the conventional systems for the $\delta^{15}\text{N}$ means ($\alpha = 0.05$, Dunnnett test). The probability of overlapping $\delta^{15}\text{N}$ values among organic and conventional system samples was calculated as 100 %. However, $\delta^{15}\text{N}$ means from hydroponic farms were statistically different from both organic and conventional systems. Surprisingly, one hydroponic system (Hydrop B) showed the highest $\delta^{15}\text{N}$ mean of lettuce, while the Hydrop A showed the lowest mean (Fig. 4). This result suggests a buffering effect of the organic substrate on N forms in the rooting medium. It may be hypothesized that mineral N from fertigation enhances N mineralization

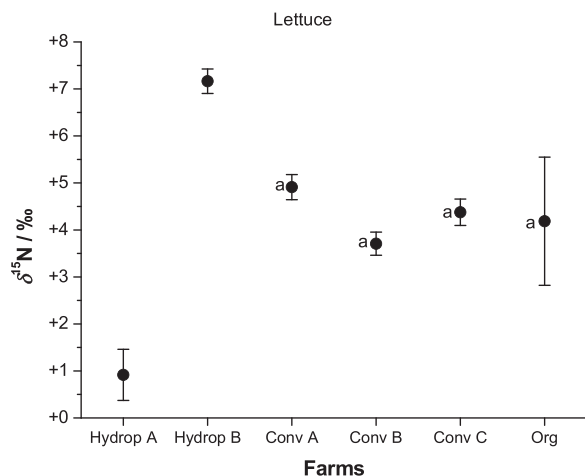


Fig. 4. $\delta^{15}\text{N}$ values (‰) of lettuce from different cropping systems in Southern Brazil. Hydrop A = Sand + fertigation; Hydrop B = Organic substrate + fertigation; Conv A and B = synthetic fertilizer only; Conv C = synthetic + organic fertilizers; Org = organic fertilizers only. Data are presented as means and standard deviations ($n = 3$). Same letter (a) represents no significant difference against Org mean (Dunnnett-test, $\alpha = 0.05$).

of the organic substrate, which, in turn, supplies ^{15}N to be preferentially absorbed by roots (Table 4).

This high $\delta^{15}\text{N}$ values ($+7.2 \pm 0.3$ ‰) of hydroponic vegetables has not been reported in the literature (Inácio et al., 2015a, 2015b). In contrast to published data (Inácio et al., 2015a, 2015b), our data show a strong limitation of $\delta^{15}\text{N}$ values in differentiating organic and conventional lettuce crops. For instance, Bateman et al. (2007) found $\delta^{15}\text{N}$ mean = $+7.6 \pm 4.1$ and $+2.9 \pm 4.3$ ‰ for organic and conventional lettuce (farm and retail sampling), respectively.

The robustness of differentiation between organic and conventional plant products depends on the difference between the $\delta^{15}\text{N}$ values of N sources (Šturm et al., 2011) used in each farming system, i.e., organic, conventional or hydroponic. Furthermore, the data suggest that the plant recovery of N from synthetic fertilizers might have been very low on some conventional farms and also in the hydroponic system. Therefore, the $\delta^{15}\text{N}$ values of plants may resemble the soil $\delta^{15}\text{N}$ or any other N source, such as substrates or cattle manure, instead of the $\delta^{15}\text{N}$ of the synthetic N-fertilizer. On the other hand, the lower the total soil-N pool, the higher the influence of N-fertilizer on $\delta^{15}\text{N}$ value during the growth of vegetables.

The effect of isotope fractionation during N transformation in soil may influence the vegetables' $\delta^{15}\text{N}$ values as a factor that drives the overlapping values. It is well known that crop $\delta^{15}\text{N}$ values vary in a wider and more positive range than their N sources (i.e. synthetic fertilizers) due to ^{15}N enrichment of the N source caused by N losses (ammonia volatilization, denitrification) (Choi et al., 2017). Thus, it may be expected for crop to show higher $\delta^{15}\text{N}$ values than synthetic fertilizer applied as a result of fractionation and recovery of the ^{15}N -enriched soil-N pool. When manure or composted manure are applied to soil as N inputs, crops might show $\delta^{15}\text{N}$ values lower or similar to these organic sources. Manure and compost have low N availability for plant uptake (Chalk et al., 2012) leading to a greater influence of the soil-N pool to crop nutrition than when synthetic fertilizer is applied. In addition, ammonification and subsequent nitrification in soil of N from manure and compost lead to a crop uptake of ^{15}N -depleted mineral N (ammonium and nitrate) compared to the N source (Choi et al., 2017).

In sum, the influence of the N source on crop $\delta^{15}\text{N}$ values not only depends on its ^{15}N natural abundance but also on N availability and its dynamics in the soil (or substrate or fertigation) which can lead to high or low N losses, and greater or smaller influence of the soil-N pool influence, and consequently changes in $\delta^{15}\text{N}$ values of the N available for crop uptake.

3.4. $\delta^{15}\text{N}$ abundance in plant parts

Leaves can be 3–7‰ enriched in ^{15}N relative to roots and the difference can be higher in dry climates (Evans, 2001). Plant parts can show different $\delta^{15}\text{N}$ values and different sensitivity to $\delta^{15}\text{N}$ -fertilizer changes as shown by Wang et al. (2016) in a controlled experiment with cauliflower (*Brassica oleracea* L. var *botrytis*). Statistically significant differences in $\delta^{15}\text{N}$ values in plant parts with stems > leaves > grains > roots for maize and sweet pepper (Choi et al., 2002; del Amor et al., 2008; Flores et al., 2007). Therefore, differences in $\delta^{15}\text{N}$ values of plant parts of grain crops or growing vegetables are expected.

In tomato plants, ripe fruit and leaves showed similar $\delta^{15}\text{N}$ means (t-test, $\alpha = 0.05$), except for conventional system B, while petiole $\delta^{15}\text{N}$ values were similar to ripe fruit. However, a pattern for unripe fruit was not found (Fig. 5). Therefore, we consider that tomato leaves could substitute ripe fruit in a field inspection for a formal certification process. In that case, a leaf sample is lighter, easier to dry and store than tomato fruit. Similar, Wang et al. (2016) pointed out that leaves and flowers of cauliflowers leaves and flowers of cauliflower were found positive correlated with $\delta^{15}\text{N}$ values of fertilizer specially and organic fertilizer were partially or totally substituted by soluble fertilizers, although, flowers were more sensitive than leaves to these changes of fertilizer.

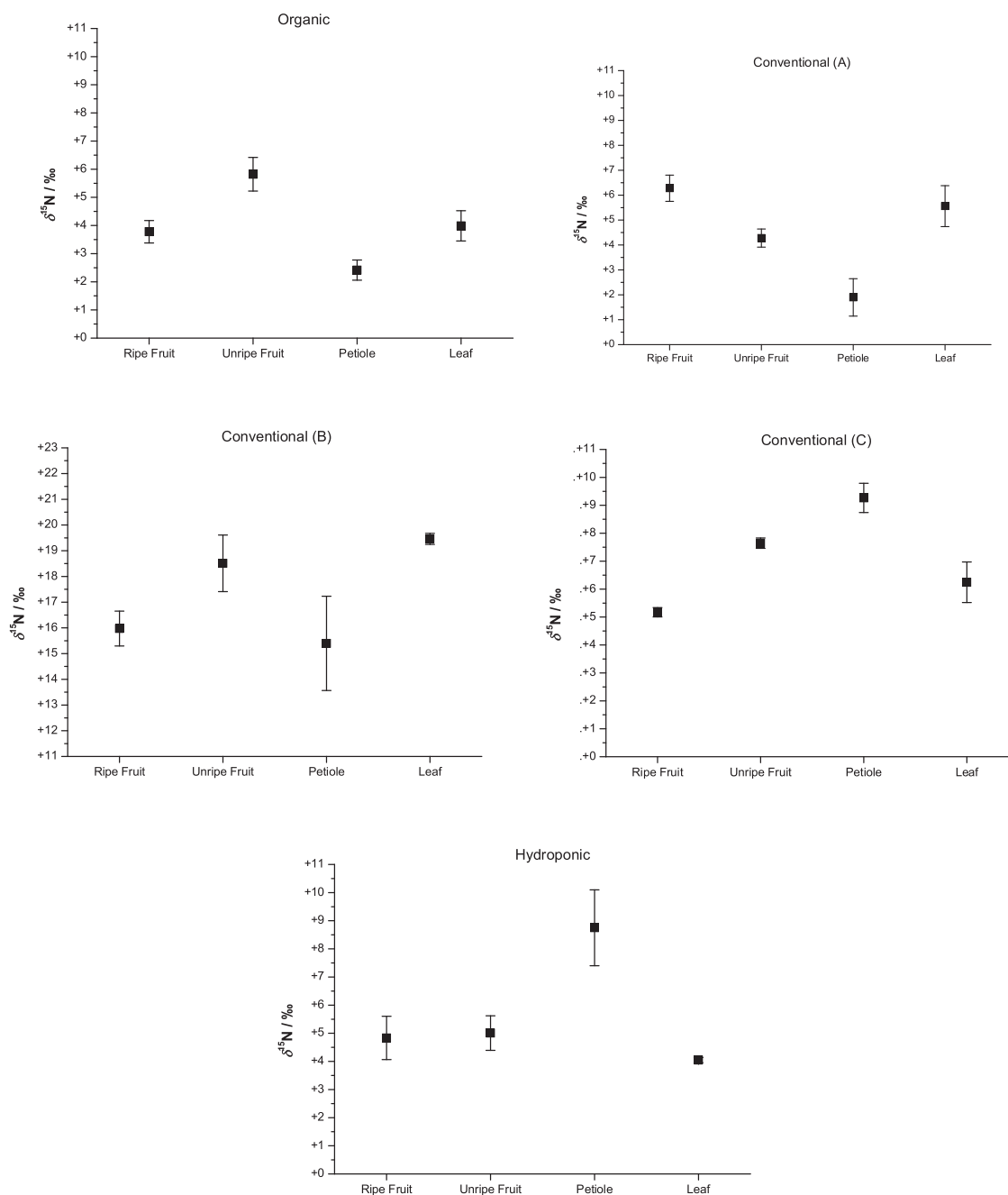


Fig. 5. Mean and standard deviation ($n = 3$) of $\delta^{15}\text{N}$ values of tomato parts (ripe fruit, unripe fruit, petiole and leaf) from different farming systems (organic, conventional and hydroponic) on five farms located in Southern Brazil.

Strawberry plant parts did not show consistent patterns of $\delta^{15}\text{N}$ means in ripe fruit (Fig. 6). For instance, in organic and hydroponic B there was no statistical difference between $\delta^{15}\text{N}$ means of plant parts. Meanwhile, conventional B showed all mean values statistically different from ripe fruit (t-test, $\alpha = 0.05$). In fact, organic and hydroponic farm B showed higher variance of plant part data compared to the other systems which contributed to the lack of separation after the means test. The influence of different N sources to strawberry plant parts was not clear. A controlled experiment using increasing levels of different types of fertilizers (organic vs. synthetic) is recommended to clarify internal fractionation patterns among plant parts and eliminate confounding factors, such timing of fertilization (Yun and Ro, 2009), which may have influenced our study.

4. Conclusion

The organic, conventional and hydroponic vegetable cropping systems selected in this study had a variety of N-inputs (substrates, organic and synthetic fertilizers) with very similar $\delta^{15}\text{N}$ values, resulting in an expressive overlapping did not allow farms to be grouped according to the ^{15}N natural abundance of N-sources. Also, soil samples from conventional farms showed higher $\delta^{15}\text{N}$ values than those from organic systems. As a consequence, $\delta^{15}\text{N}$ values as a tool to differentiate plant products from different farming systems had limited value. However, it may be a potentially more useful technique for strawberries but not for tomatoes or lettuce. The inconsistency found in the differentiation between farming systems can be attributed to the overlapping $\delta^{15}\text{N}$ values of the N sources used on those farms. The unknown magnitude of isotope fractionation occurring in every farming system is a secondary

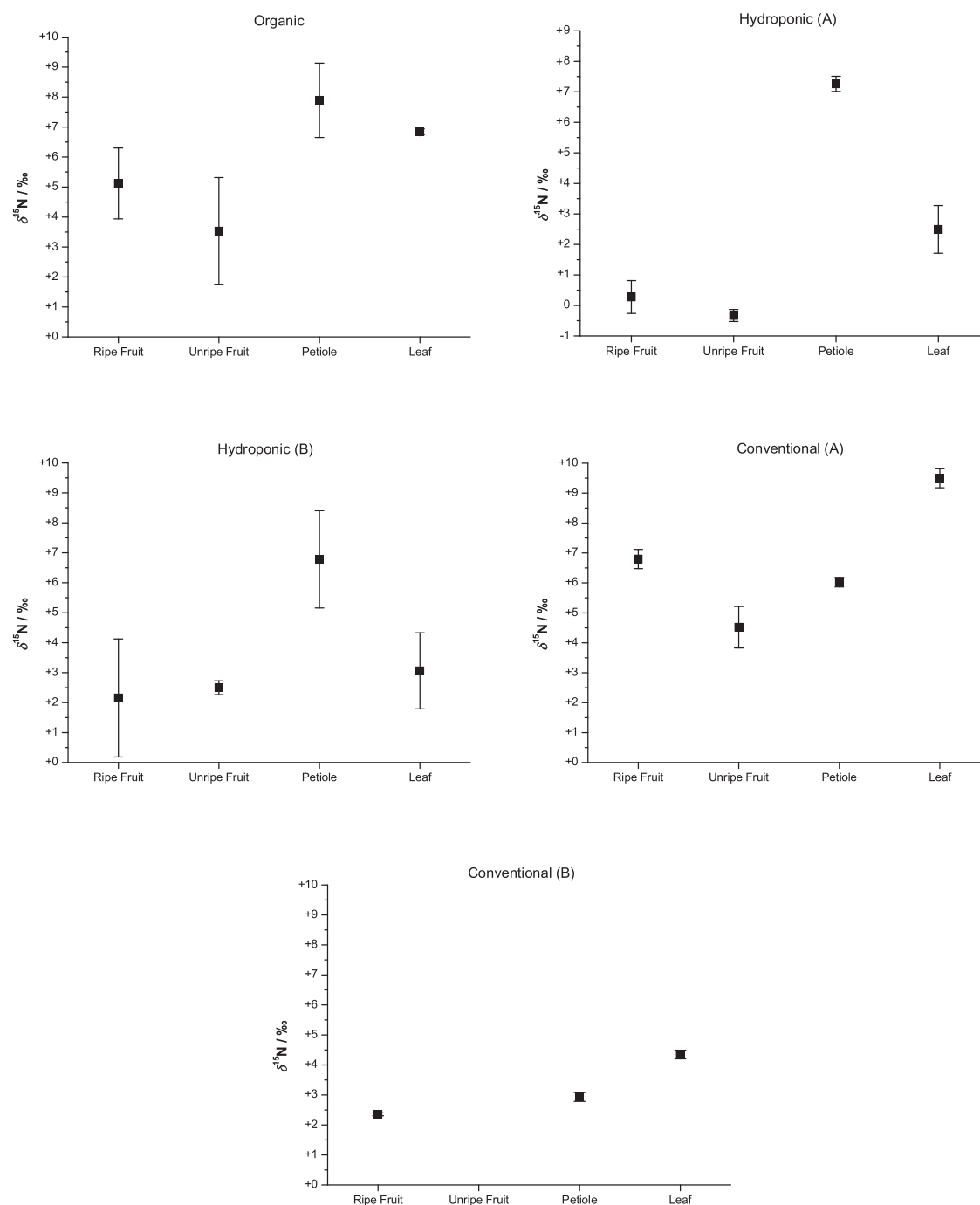


Fig. 6. Mean and standard deviation ($n = 3$) of $\delta^{15}\text{N}$ values of strawberry parts (ripe fruit, unripe fruit, petiole and leaf) from different cropping systems (organic, conventional and hydroponic) on five farms located in Southern Brazil.

influencing factor that may be the subject of future studies. Another confounding factor is the use of organic N sources on conventional farms. Regarding plant parts, we consider that leaves of tomato plants could substitute ripe fruit in a field inspection for formal certification. However, for strawberries there was no similar consistency. Our findings suggest that more accurate studies must be carried out in Brazil before recommending crop $\delta^{15}\text{N}$ value as a tool to differentiate vegetable products according to their fertilization regimes.

CRediT authorship contribution statement

Caio de Teves Inácio: Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration,

Resources, Visualization, Writing - original draft, Writing - review & editing. **Járisson Cavalcante Nunes:** Data curation, Visualization, Conceptualization, Writing - original draft. **Jerônimo Luiz Andriolo:** Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Visualization, Writing - original draft, Writing - review & editing. **Phillip Michael Chalk:** Data curation, Visualization, Conceptualization, Writing - original draft. **Sandro José Giacomini:** Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Visualization, Writing - original draft, Writing - review & editing. **Lessandro De Conti:** Investigation, Software, Formal analysis, Supervision, Writing - review & editing. **Amanda Veridiana Krug:** Investigation, Software, Formal analysis, Supervision, Writing - review

& editing. **Betania Vahl de Paula:** Investigation, Software, Formal analysis, Supervision, Writing - review & editing. **Gustavo Brunetto:** Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Visualization, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Alvarenga, M.A.R., Souza, R.A.M., 2004. Comercialização, colheita classificação e embalagens. Tomate: Produção Em Campo, Em Casa-de-Vegetação e Em Hidroponia. UFLA, Lavras.
- Bateman, A.S., Kelly, S.D., Jickells, T.D., 2005. Nitrogen isotope relationships between crops and fertilizer: implications for using nitrogen isotope analysis as an indicator of agricultural regime. *J. Agric. Food Chem.* 53 (14), 5760–5765. <https://doi.org/10.1021/jf050374h>.
- Bateman, A.S., Kelly, S.D., Woolfe, M., 2007. Nitrogen isotope composition of organically and conventionally grown crops. *J. Agric. Food Chem.* 55, 2664–2670. <https://doi.org/10.1021/jf0627726>.
- Camin, F., Perini, M., Bontempo, L., Fabroni, S., Faedi, W., Magnani, S., Baruzzi, G., Bonoli, M., Tabilio, M.R., Musmeci, S., Rossmann, A., Kelly, S.D., Rapisarda, P., 2011. Potential isotopic and chemical markers for characterising organic fruits. *Food Chem.* 125 (3), 1072–1082. <https://doi.org/10.1016/j.foodchem.2010.09.081>.
- Cejudo, E., Schiff, S.L., 2018. Nitrogen isotope fractionation factors (α) measured and estimated from the volatilisation of ammonia from water at pH 9.2 and pH 8.5. *Isotopes Environ. Health Stud.* 54, 642–655. <https://doi.org/10.1080/10256016.2018.1526792>.
- Celano, G., Alluvione, F., Mohamed, M.A.A.A., Spaccini, R., 2012. The stable isotopes approach to study C and N sequestration processes in a plant–soil system. In: Springer (Ed.), *Carbon Sequestration in Agricultural Soils*. Springer, Berlin Heidelberg, Berlin, Heidelberg, pp. 107–144. https://doi.org/10.1007/978-3-642-23385-2_5.
- Chalk, P.M., Magalhães, A.M.T., Inácio, C.T., 2012. Towards an understanding of the dynamics of compost N in the soil-plant-atmosphere system using ^{15}N tracer. *Plant Soil* 362, 373–388. <https://doi.org/10.1007/s11104-012-1358-5>.
- Chalk, P.M., Inácio, C.T., Magalhães, A.M.T., 2014. From fertilizer to food: tracing nitrogen dynamics in conventional and organic farming systems using ^{15}N natural abundance. *Int. Symp. Manag. soils food Secur. Clim. Chang. Adapt. Mitig.*
- Chalk, P.M., Inácio, C.T., Craswell, E.T., Chen, D., 2015. On the usage of absolute (α) and relative (δ) values of ^{15}N abundance. *Soil Biol. Biochem.* 85, 51–53. <https://doi.org/10.1016/j.soilbio.2015.02.027>.
- Choi, W.-J., Lee, S.M., Ro, H.M., Kim, K.C., Yoo, S.H., 2002. Natural ^{15}N abundances of maize and soil amended with urea and composted pig manure. *Plant Soil* 245, 223–232. <https://doi.org/10.1023/A:1020475017254>.
- Choi, W.-J., Ro, H.-M., Hobbie, E.A., 2003. Patterns of natural ^{15}N in soils and plants from chemically and organically fertilized uplands. *Soil Biol. Biochem.* 35, 1493–1500. [https://doi.org/10.1016/S0038-0717\(03\)00246-3](https://doi.org/10.1016/S0038-0717(03)00246-3).
- Choi, W.-J., Arshad, M.A., Chang, S.X., Kim, T.H., 2006. Grain ^{15}N of crops applied with organic and chemical fertilizers in a four-year rotation. *Plant Soil* 284, 165–174. <https://doi.org/10.1007/s11104-006-0038-8>.
- Choi, W.-J., Kwak, J.-H., Lim, S.-S., Park, H.-J., Chang, S.X., Lee, S.-M., Arshad, M.A., Yun, S.-I., Kim, H.-Y., 2017. Synthetic fertilizer and livestock manure differently affect $\delta^{15}\text{N}$ in the agricultural landscape: a review. *Agric. Ecosyst. Environ.* 237, 1–15. <https://doi.org/10.1016/j.agee.2016.12.020>.
- Coplen, T.B., Böhlke, J.K., De Bièvre, P., Ding, T., Holden, N.E., Hopple, J.A., Krouse, H.R., Lamberty, A., Peiser, H.S., Revesz, K., Rieder, S.E., Rosman, K.J.R., Roth, E., Taylor, P.D.P., Vocke, R.D., Xiao, Y.K., 2002. Isotope-abundance variations of selected elements (IUPAC Technical Report). *Pure Appl. Chem.* 74, 1987–2017. <https://doi.org/10.1351/pac200274101987>.
- Danezis, G.P., Tsagkaris, A.S., Camin, F., Brusica, V., Georgiou, C.A., 2016. Food authentication: techniques, trends & emerging approaches. *TrAC - Trends Anal. Chem.* 85, 123–132. <https://doi.org/10.1016/j.trac.2016.02.026>.
- Dawson, T.E., Mambelli, S., Plamboeck, A.H., Templer, P.H., Tu, K.P., 2002. Stable C and N isotopes in ecology and environmental science. *Rev. Ecol. Evol. Syst.* <https://doi.org/10.1146/annurev.ecolsys.33.020602.095451>.
- del Amor, F.M., Navarro, J., Aparicio, P.M., 2008. Isotopic discrimination as a tool for organic farming certification in sweet pepper. *J. Environ. Qual.* 37, 182. <https://doi.org/10.2134/jeq2007.0329>.
- Evans, R.D., 2001. Physiological mechanisms influencing plant nitrogen isotope composition. *Trends Plant Sci.* 6, 121–126. [https://doi.org/10.1016/S1360-1385\(01\)01889-1](https://doi.org/10.1016/S1360-1385(01)01889-1).
- Flores, P., Fenoll, J., Hellín, P., 2007. The feasibility of using $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values for discriminating between conventionally and organically fertilized pepper (*Capsicum annuum* L.). *J. Agric. Food Chem.* 55, 5740–5745. <https://doi.org/10.1021/jf0701180>.
- Hristov, A.N., Zaman, S., Vander Pol, M., Ndegwa, P., Campbell, L., Silva, S., 2009. Nitrogen losses from dairy manure estimated through nitrogen mass balance and chemical markers. *J. Environ. Qual.* 38, 2438. <https://doi.org/10.2134/jeq2009.0057>.
- Inácio, C.T., Chalk, P.M., 2017. Principles and limitations of stable isotopes in differentiating organic and conventional foodstuffs: 2. Animal products. *Crit. Rev. Food Sci. Nutr.* 57 (1), 181–196. <https://doi.org/10.1080/10408398.2014.887056>.
- Inácio, C.T., Chalk, P.M., Magalhães, A.M.T., 2015a. Principles and limitations of stable isotopes in differentiating organic and conventional foodstuffs: 1. Plant products. *Crit. Rev. Food Sci. Nutr.* 55, 1206–1218. <https://doi.org/10.1080/10408398.2012.689380>.
- Inácio, C.T., Urquiaga, S., Chalk, P.M., Mata, M.G.F., Souza, P.O., 2015b. Identifying N fertilizer regime and vegetable production system in tropical Brazil using ^{15}N natural abundance. *J. Sci. Food Agric.* 95, 3025–3032. <https://doi.org/10.1002/jsfa.7177>.
- Inácio, C.T., Magalhães, A.M.T., Souza, P.O., Chalk, P.M., Urquiaga, S., 2018. The relative isotopic abundance ($\delta^{13}\text{C}$, $\delta^{15}\text{N}$) during composting of agricultural wastes in relation to compost quality and feedstock. *Isotopes Environ. Health Stud.* 54, 185–195. <https://doi.org/10.1080/10256016.2017.1377196>.
- Kim, Y.-J., Choi, W.-J., Lim, S.-S., Kwak, J.-H., Chang, S.X., Kim, H.-Y., Yoon, K.-S., Ro, H.-M., 2008. Changes in nitrogen isotopic compositions during composting of cattle feedlot manure: effects of bedding material type. *Bioresour. Technol.* 99, 5452–5458. <https://doi.org/10.1016/j.biortech.2007.11.012>.
- Lee, C., Hristov, A.N., Cassidy, T., Heyler, K., 2011. Nitrogen isotope fractionation and origin of ammonia nitrogen volatilized from cattle manure in simulated storage. *Atmosphere (Basel)* 2 (3), 256–270. <https://doi.org/10.3390/atmos2030256>.
- Martinelli, L.A., Piccolo, M.C., Townsend, A.R., Cuevas, E., Robertson, G.P., Treseder, K., 1999. Martinelli 1999.pdf. pp. 45–65. <https://doi.org/10.1023/A:1006100128782>.
- Nakano, A., Uehara, Y., Yamauchi, A., 2003. Effect of organic and inorganic fertigation on yields, $\delta^{15}\text{N}$ values, and $\delta^{13}\text{C}$ values of tomato (*Lycopersicon esculentum* Mill. cv. Saturn). *Plant Soil* 255 (1), 343–349. <https://doi.org/10.1023/A:1026180700963>.
- Oberson, A., Nanzer, S., Bosshard, C., Dubois, D., Mäder, P., Frossard, E., 2007. Symbiotic N_2 fixation by soybean in organic and conventional cropping systems estimated by ^{15}N dilution and ^{15}N natural abundance. *Plant Soil* 290, 69–83. <https://doi.org/10.1007/s11104-006-9122-3>.
- Rogers, K.M., 2008. Nitrogen isotopes as a screening tool to determine the growing regimen of some organic and nonorganic supermarket produce from New Zealand. *J. Agric. Food Chem.* 56 (1), 4078–4083. <https://doi.org/10.1021/jf800797w>.
- Schmidt, H., Roßmann, A., Voerkelius, S., Schnitzler, W.H., Georgi, M., Graßmann, J., Zimmermann, G., Winkler, R., 2005. Isotope characteristics of vegetables and wheat from conventional and organic production. *Isotopes Environ. Health Stud.* 41, 223–228. <https://doi.org/10.1080/10256010500230072>.
- Šturim, M., Kacjan-Maršić, N., Lojen, S., 2011. Can $\delta^{15}\text{N}$ in lettuce tissues reveal the use of synthetic nitrogen fertiliser in organic production? *J. Sci. Food Agric.* 91 (2), 262–267. <https://doi.org/10.1002/jsfa.4179>.
- Wang, H., Hu, G., Lou, Y., Su, Z., Zhuge, Y., Meng, F., 2016. Responses of natural ^{15}N abundance in cauliflower (*Brassica oleracea* L. var. botrytis) and soil to the application of organic and chemical fertilizers. *Can. J. Plant Sci.* 96, 819–827. <https://doi.org/10.1139/cjps-2015-0365>.
- Yun, S.-I., Ro, H.-M., 2008. Stable C and N isotopes: a tool to interpret interacting environmental stresses on soil and plant. *J. Appl. Biol. Chem.* 51, 262–271. <https://doi.org/10.3839/jabc.2008.041>.
- Yun, S.-I., Ro, H.-M., 2009. Natural ^{15}N abundance of plant and soil inorganic-N as evidence for over-fertilization with compost. *Soil Biol. Biochem.* 41, 1541–1547.